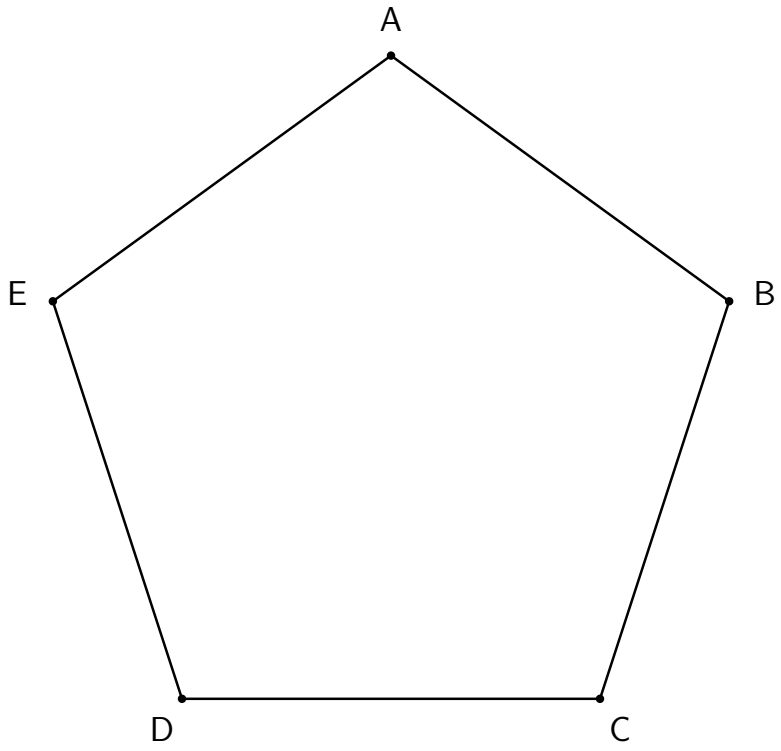
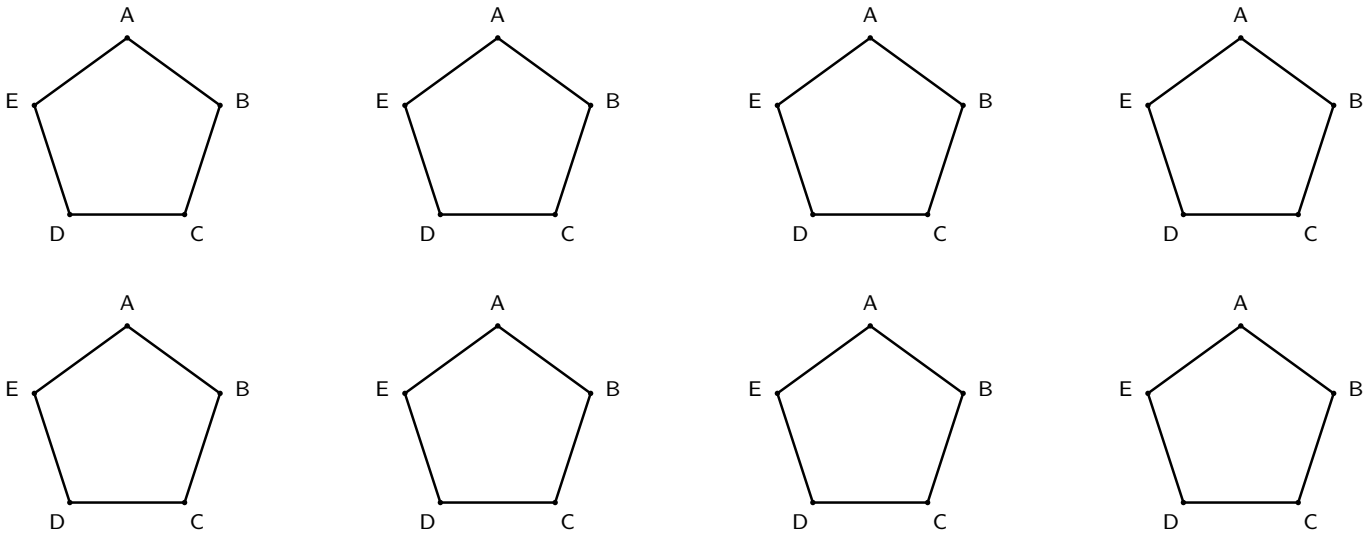
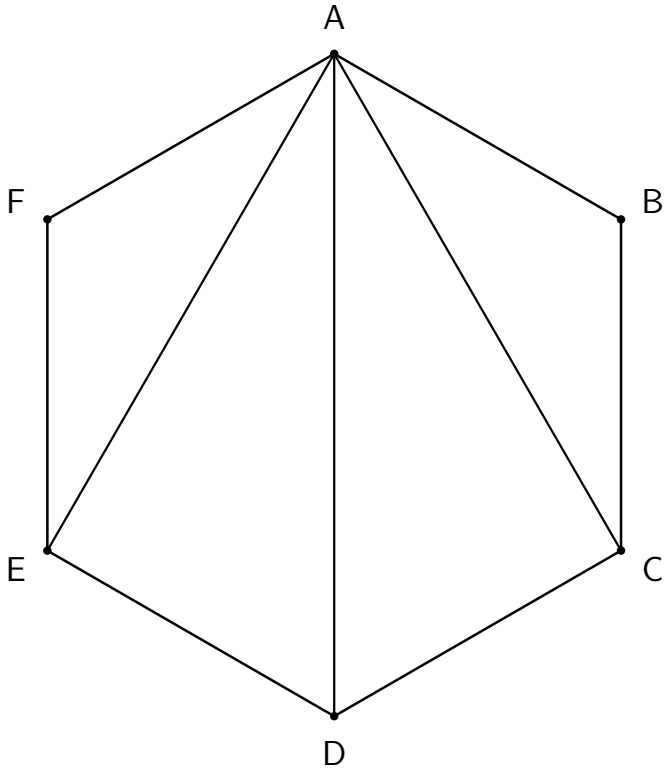
Fill the whole polygon with triangles, using straight lines between printed corners. Lines inside a polygon may meet at corners but may not cross. Keep the corner labels fixed, even if you turn the page. Use tracing paper to change printed lines.

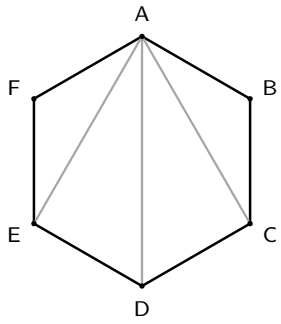
Problem 1: Find every filling of the five-corner polygon. Explain how your collection shows that none are missing.

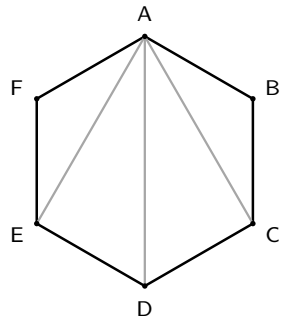


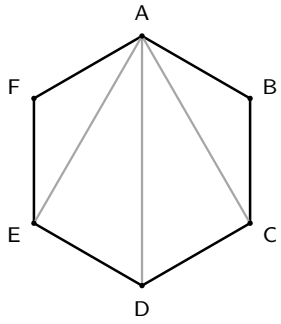


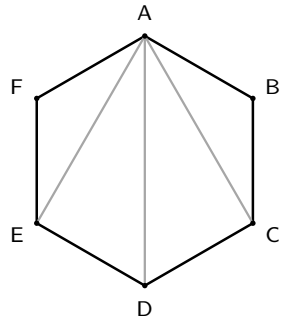
Problem 2: Erase one inside line and replace it with a different line so the polygon stays filled with triangles. Find every result of one such change from each drawing.

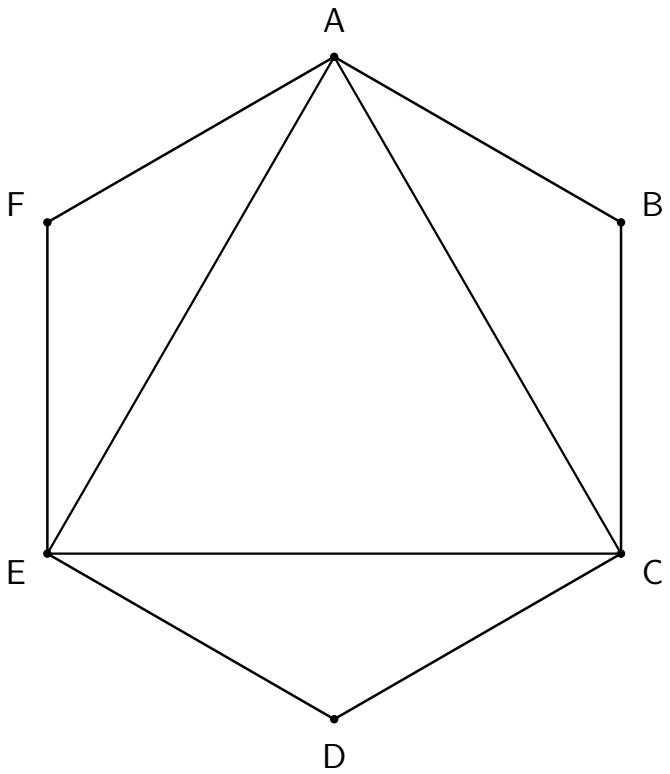


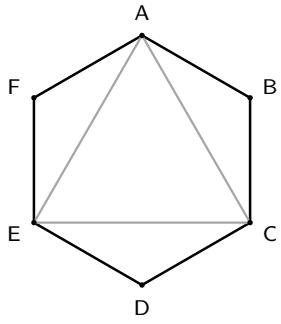


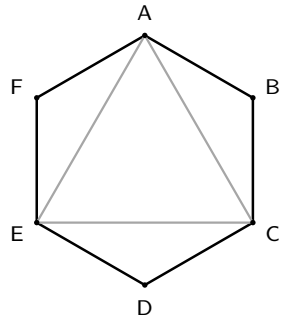


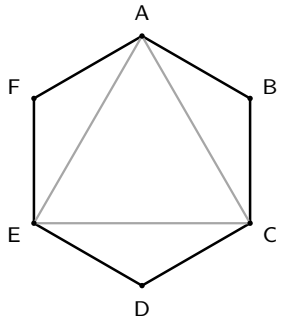


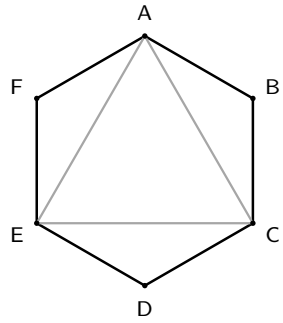




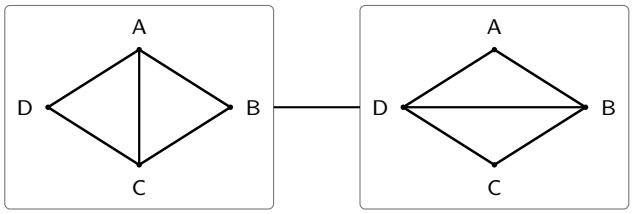




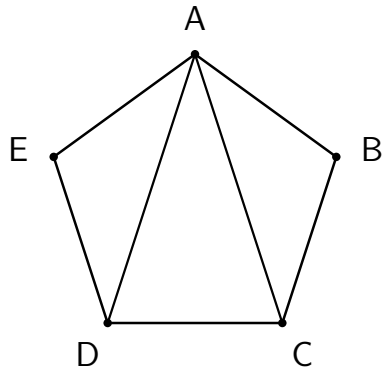


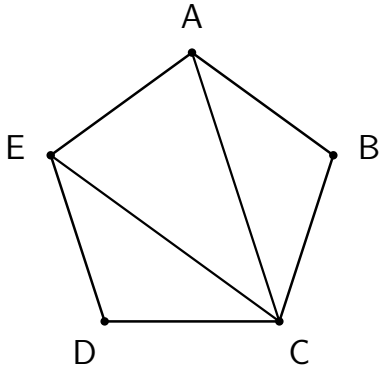


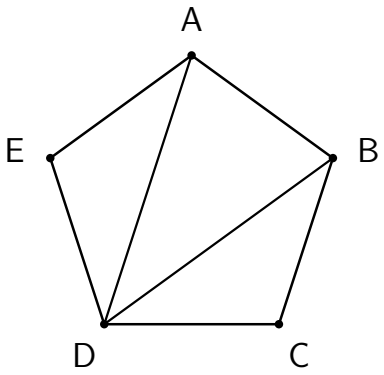
Problem 3: A change of one inside line that keeps the polygon filled with triangles is called a flip. Join two drawings when one flip changes one into the other. Include every possible join. Can you return to a filling after an odd number of flips? Show a route, or explain why it is impossible.

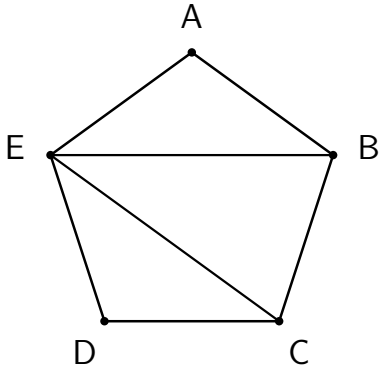


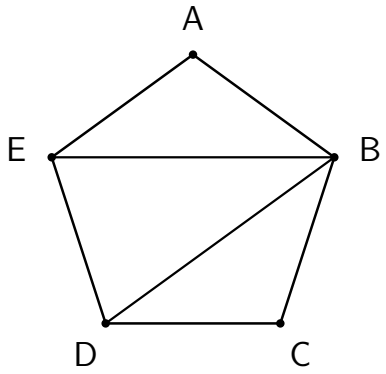
Example: erase AC, draw BD. The join connects whole fillings; it is not an inside line.



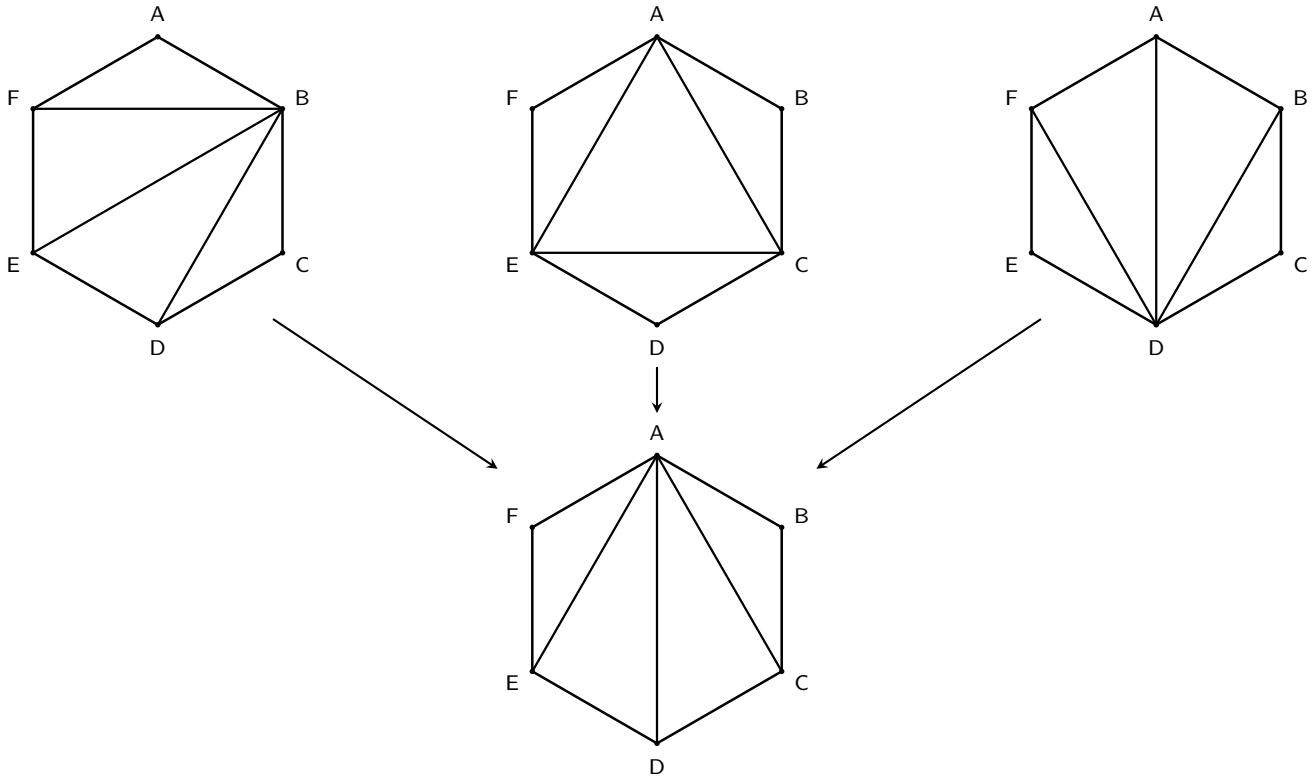


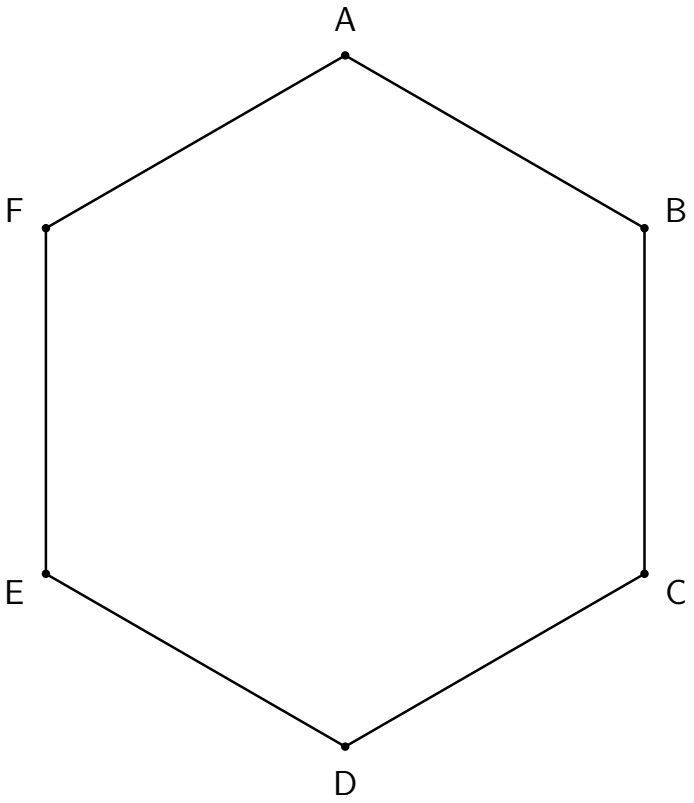






Problem 4: Change each of the three starting fillings into the fourth filling using as few flips as possible. Explain why no shorter routes can work.



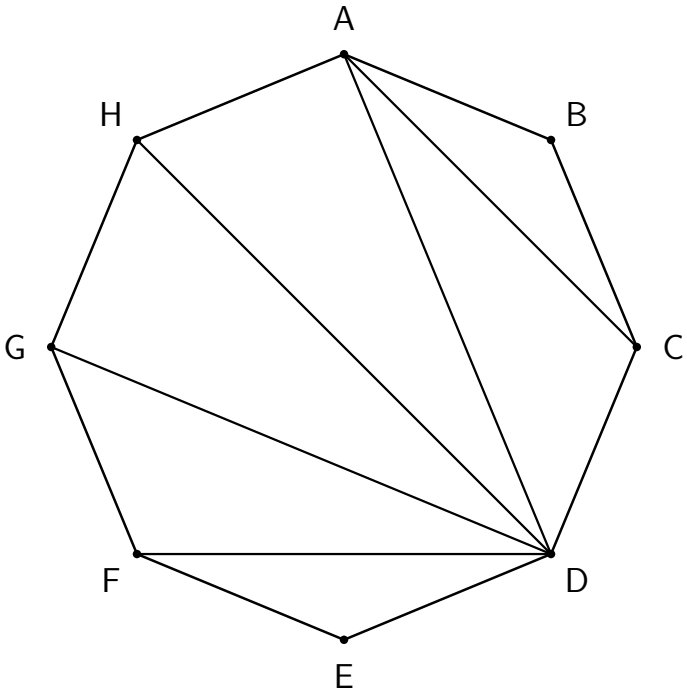


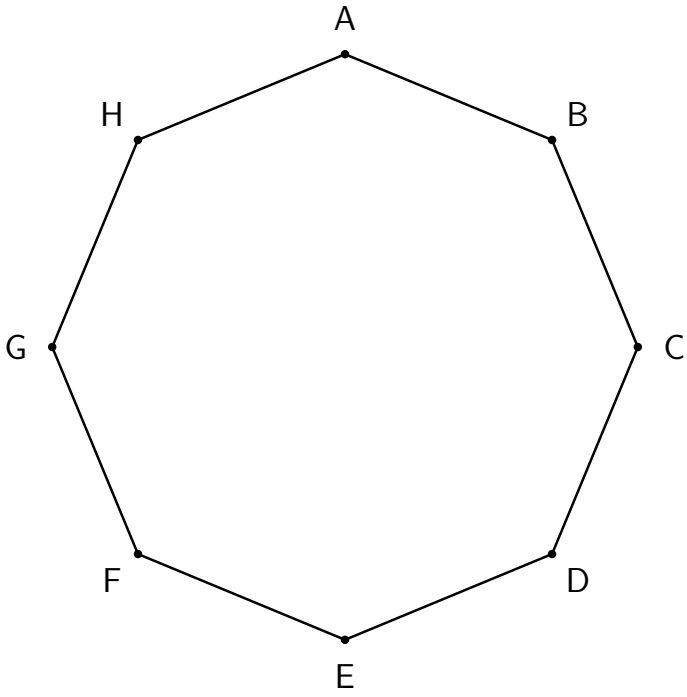
Left start

Middle start

Right start

Problem 5: A filling whose inside lines all meet at one corner is called a fan. Find the fewest flips from this filling to a fan at A, and to a fan at E. Explain why each route is shortest.

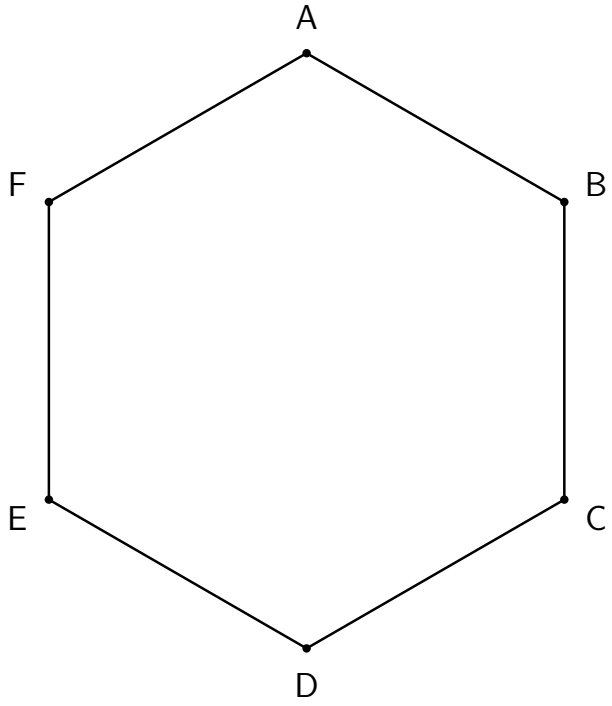


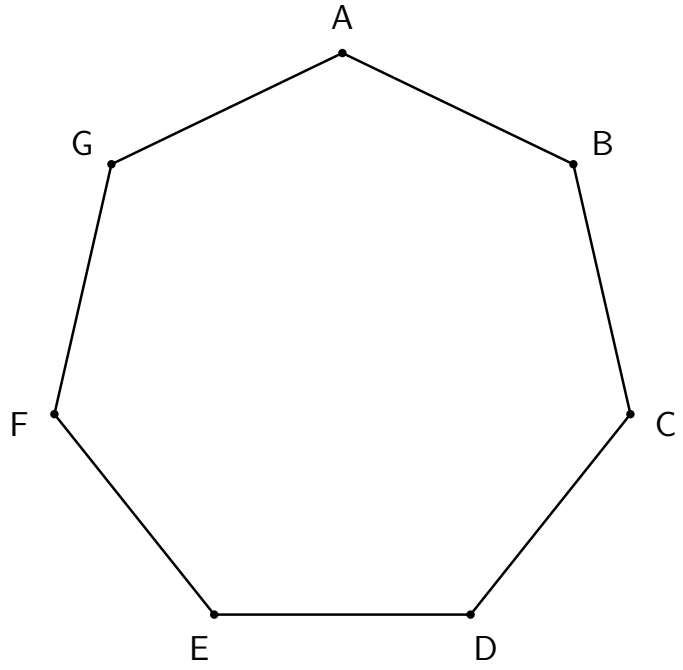


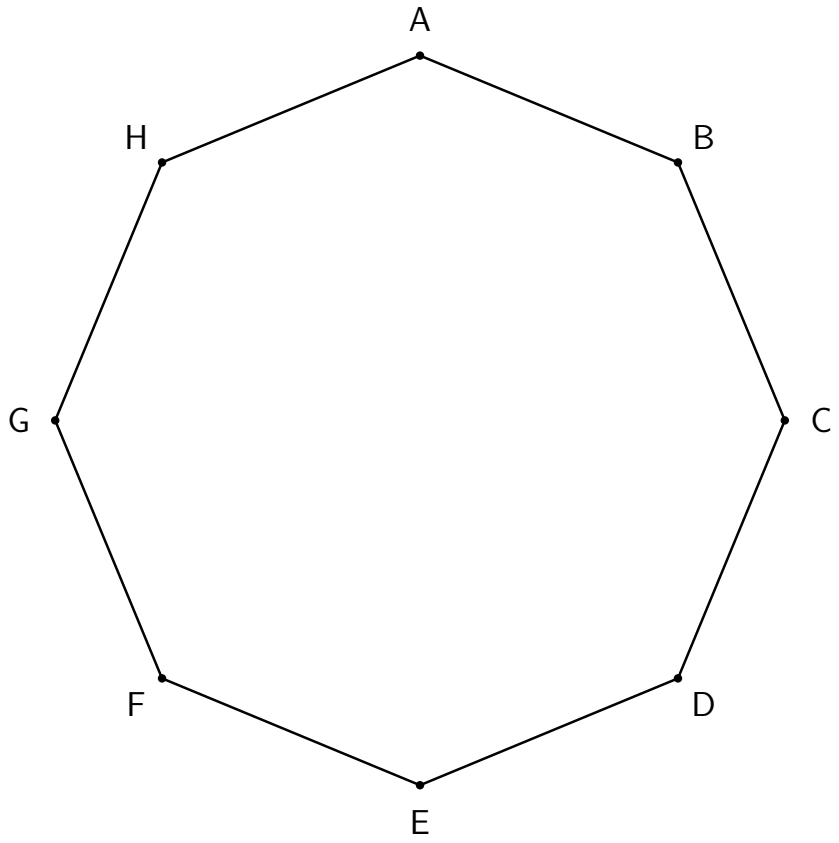
To a fan at A

To a fan at E

Problem 6: Find a rule for the fewest flips needed to turn any filling into a fan at A. Explain why your rule works for every starting filling and every number of corners, when all corners point outward.







Problem 7: Find a route of flips between these two fillings. Can any two fillings of the same polygon be joined by flips? Explain why your answer works for any polygon with no inward-pointing corners.

